Samuel Anes

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EDUCATION

Carnegie Mellon University — Pittsburgh, PA
Bachelor of Science in Electrical and Computer Engineering — Graduating May 2030
Selected Coursework: Intro to Electrical and Computer Engineering, Calculus in Three Dimensions, Concepts of Math

Homeschool, Stanford Online High School (Graduated June 2026) — Sudbury, MA
GPA: Weighted: 4.4 / Unweighted: 3.97 — SAT: 1530 ACT: 35
Selected Coursework: AP Physics C: Electricity & Magnetism, AP Physics C: Mechanics, Linear Algebra, Harvard CS50x

SKILLS

Programming: Python, Java, C, C++, SQL, JavaScript, HTML, CSS, Svelte
Software & Tools: GitHub, Claude Code, ROS2, Verilog, Linux, Android Studio, OnShape
Hardware: Arduino, 3D Printers, CNC Machines, Multimeter, Oscilloscope, LCR Meter

ENGINEERING EXPERIENCE

Carnegie Mellon Racing - Formula SAE Electric (September 2026-Present) — Pittsburgh, PA
Delaunay Edge Selection Lead, Driverless Path Planning — October 2026 -Present

- Bringing up a naive R&D algorithm into a robust competition-ready system. Implements a cost function on the edges produced by a Delaunay Triangulation algorithm to determine the correct path through the track given a set of colorless cones. The car now plans 54% further ahead (8.0 → 12.3 m) inside a 10 Hz loop.

Regenerative Braking Lead, Dynamics Controls — September 2026-Present

- Designed three unique regenerative braking algorithms, two using one pedal and one using two pedals. Proposed the algorithms to the team, then tested the two-pedal algorithm on the car.
- To determine the efficacy of regenerative braking, I fitted a braking model on runs with regen off, measured what was left over with it on, and got 5–7 N·m per m/s^2. Currently validating the model on driverless runs which only use regenerative braking. Next steps include characterizing thermal impact of regen braking.

FIRST Tech Challenge Team #2845 (September 2022-April 2026) — Tampa, FL
Team Captain | FIRST Dean's List Semi-Finalist — September 2024-April 2026

- Rebuilt the team's Java software architecture into a modular command-based framework supporting sequential and parallel execution, enabling concurrent subsystem control during autonomous routines.
- Developed three-wheel + IMU odometry with periodic computer vision sensor fusion corrections and integrated the localization with a PID-based point-to-point robot navigation for autonomous field movement.
- Developed OpenCV computer-vision pipelines using HSV color segmentation, contour filtering, and geometric constraints to detect game elements and estimate their position relative to the robot.
- Developed a data-driven launcher velocity model by regressing target distance against measured AprilTag image area, automatically selecting launch velocity for autonomous scoring across varying distances.

Sabacc: Open Source Multiplayer Full-Stack Web Application
Project Founder and Leader — April 2024-Present

- Built and deployed a real-time multiplayer web application serving 1,000+ users, using Svelte, Flask, SQL, and WebSockets for reactive UI, backend game logic, persistence, and low-latency gameplay.
- Architected an object-oriented backend supporting multiple game variants through shared abstractions, and migrated between SQLite and PostgreSQL while maintaining persistent game data.

EXTRA

Eagle Scout, Scouting America — October 2026
Amateur Extra, FCC Licensed Amateur Radio Operator — July 2023